L9.5: Continuity for multivariable functions

Transcript Language

English

Hello and welcome to the Maths 2 component of the online BSc programme on data science

and programming. In this video we are going to talk about continuity for multivariable

functions.

Let us recall that in the previous video, we have studied the notion of limits for sequences

and limits for scalar valued multivariable functions. So, these were sequences in higher dimensions. So, let an tilde be a sequence in RP denote the coordinates of an tilde as

an1, an2 up to anp. We say an tilde is limit a tilde is a1, a2 ap if the sequence in the

ith coordinate has limit ai.

So, which means that if you take the sequences given by each coordinate that converges to

some number ai, and you put those numbers into this vector, and then we say an tilde

has that element. So, ani in particular has to converge to ai for each i. So, if each

of those coordinate limits exists, then the sequence an tilde as a limit, if even one

of those fails to exist, it does not have a limit, so limit will not exist.

So, we use this idea to define limits for functions at a point. So, let f be a scalar

valued multivariable function defined on a domain D in our case is that you have a sequence

converging to a tilde. So, that sequence must belong to D. So, if there exists a real number

L such that f of an tilde tends to L for all sequences an tilde and this is the important part for all sequences an tilde such that an tilde tense to a tilde, then we say that

the limit of f at a tilde exists n equals L.

And we have seen examples of these things. So, we denote this by limit X tilde tends

to a tilde f of x tilde is L. So, in the previous video, we also studied some properties, and

using them how we can find these limits. So, for example, for polynomials for rational

functions and so on. We also saw an example where substitution directly does not work.

And so, we have to be careful when we find these limits.

Let us first define what is the limit of a vector valued function at a point. So, we

have seen for a scalar valued function what happens, so vector valued function is a vector

of functions. And so, what we basically will do is demand that for each of them, we study

what happens.

So, let f D to Rm be a vector valued multivariable function defined on a domain D in Rk, a tilde will be a point such that there exists a sequence in D which converges to a tilde if fi is the

ith component function of the function f, and in a minute, we will see examples of what

I mean. Then fi is a scalar valued function from D to R. And then suppose for each i the

limit X tilde tends to a tilde fi of X tilde exists and equals Li. Then we define L tilde

to be L1, L2, Lm then limit X tilde tends a tilde f of X tildes is L tilde.

So, this is not the definition, but this is what I would call a working definition. And

it is equivalent to the original definition, which we would not talk about. So, this is

equivalent to as X tilde comes closer and closer to a tilde f of X tilde comes closer

and closer to this vector L tilde. Remember now, that f is a vector valued function, so,

it will take vectors as values. And so it has to come closer and closer to some vector

L tilde.

And if for some i, the limit a tilde for fi does not exist, then the limit of f at a tilde will does not exist. So, we have seen this idea already for sequences. For sequences,

what did we do, we took each coordinate sequence, and then check for the limit for that. The

same thing is happening here for functions. If you have vector valued functions, you take

each component and check for the limit.

So, let us do a couple of examples. So, here is limit X tildes tends to 1 comma 2, x squared

y plus y cubed comma e to the power xy comma x squared minus 1 by y cubed minus 2. So,

recall from a previous video that if we had polynomials, we can just substitute. So, the

first so this first one exists, so I will write the what is equal to in a minute. So,

to check this limit, what do I have to do I have to take each coordinate function, what

are the coordinate functions, this is the first coordinate function f1, this is the

second coordinate function f2, this is the third coordinate function f3.

And then I have asked, let us look at these three functions, what happens for each of these limits? Each of these exists, then you just put that vector together. And that is

what this limit is. Even one of them does not exist, this one will not exist. So, for

the first one, we have a polynomial. So, we can just substitute, that is what you saw

on the in the previous video. So, this is 1 squared times 2 plus 2 cubed, that is 10.

For the second one, we saw that e to the 4x y also you can substitute by using that xy

is a polynomial and then using the fact that you are composing with the exponential function.

So, this follows from the composition property. So, this is e to the power 1 times 2, so this

is E squared. And for the third one, you use the quotient. So, let us check that the quotient

is. Well, the main problem comes from the denominator. So, the denominator here, the

limit is 2 cube minus 2, which is non-zero. So, no problem we can just substitute. This

is one squared minus 1 by 2 cubed minus 2, this is 0.

So, each of these limits exists. And now you just take these components and put them in their proper place. So, this is the limit. So, I hope it is clear what now the previous

slide said, you take each component function check for its limit, once it exists, the function

for this and the limit for this entire function exists.

Let us look at the second one. So, this is the first component function. This is the

second component function. So, for the first component function, you have sin xy divided

by xy limit xy tend to 00. So, I want to check what happens to this limit. Well, as we noted

in the previous video, if you have things which are not like polynomial rational functions,

the way to do them is to try and use the composition law.

So here, this is a composition of two functions. The first function is xy. The second function

is sin of u. So, as xy tends to 00, this to 0, and as u tends to 0, my bad, this is not

sin of u, this is sin of u by u. And as u tends to 0 sin of u by u tends to 1, so this

limit is going to be 1, let us see what happens to the second coordinate. So, is x cube minus y squared x divided by x squared plus y squared squared.

But in the previous video, we have seen at the end that this limit actually does not

exist. So, I suggest you recall this. And so, one of these coordinate functions does

not have a limit that means this also does not exist. So, I hope it is very clear what

it means for vector valued functions to have limits, basically it boils down to asking

for scalar valued functions, what is what the limits are, which are each component fine.

So, what is the limit of a function at a point along a curve? So, now we have seen examples

of or rules about how to compute the limits of functions at a point that is for scalar

valued functions that are at a point if you can write them in terms of other functions

that you already know have limits. So, addition, subtraction, products composition quotients

if the denominator is nice, so these are these are ways we know of how to prove that a limit

actually does exist. How do we know that the limit does not exist?

So, we have seen the idea of how to do this in the previous video at the end, when we

looked at an example that we just saw in the previous slide. And now we are going to refine

that idea. And to do that we are going to talk about the limit of a function at a point

along a curve. And this will really generalise what happens in one variable calculus.

So, let f be a scalar valued multivariable function defined on a domain D in Rk and a

tilde will be a point such that there exists a sequence in D which converges to a tilde.

This is our standard hypothesis. Because without this, we do not know how to make sense of

limits at that point a tilde. Let C be a curve passing through the point a tilde belonging

to the domain D.

So, recall that we have studied curves in one variable calculus, so a curve is nothing

but something like a wavy line may be something, it could be a circle, straight lines are special

cases of curves, but curves can have curvature, they can be curve. So, we want to look at curves, which are passing through that point a tilde and which also belong to the domain

D. Except for that point a tilde, a tilde could belong may not belong.

The limit of f at a tilde the curve C, I will repeat that along the curve C exists and equals

L if for every sequence an tilde contained in C, which converges to a tilde, the sequence

af f of an tilde converges to L. So, now what we are doing is, we are saying well, I want

to only check for what happens along this curve. I do not for now, I do not care about

the function may be defined on the entire, let us say R2, but I only want to see what

happens to this function on this particular curve.

We have seen this kind of idea, before, when we talked about directional derivatives or

partial derivatives, we restricted our attention to a particular line, and then said what happens

to the function along that line. Of course, there we talked about rate of change, here,

we just want to talk about the function itself, not we are not yet going to rate of change.

So, what happens to the function when we restrict it to a curve? So, on that, as you come closer

to the point a tilde, do the function values come close somewhere.

And if they do, and that is the number L, then we say that the limit of f at a tilde

along the curve C exists and is L. So, let us again look at that same example from two

slides ago. So, let us look at g of xy is x cubed minus y squared x divided by x squared

plus y squared squared. So, limit xy tends to 0, 0 along the x axis, let say. So, along

the x axis, the y is 0. So, this reduces to a function of one variable. So, along the

xy, to solve this, I will say so. So, I will solve this in a minute.

So, along the x axis, the y coordinate is 0. And the hence the function is g of x comma

0. So, it is a function of one variable. And that is really the point of defining this

notion of long ago. So, g of x comma 0, which is x cubed, divided by x to the power 4, which

is 1 by x. And now we know what happens. So, this limit is exactly limit x tends to 0 1 by x, and this does not exist.

So, along the curve, y is equal to 0, which is the x axis, this limit does not exist,

we could ask the same question about what happens along the y axis. So, limit x y tends

to 0, 0 g xy along y the y axis, and you can see that if you look at g of 0 comma y, then

then the numerator is 0, the denominator is y to the power 4, so this is 0. And so, this

is limit as y tends to 0 of the function 0, so this is 0.

So, for g along the curve, which is the x axis, the limit does not exist along the y

axis, it is 0, we could do some other curve, you can choose your favourite curve, let us

say I choose the curve y is equal to x. So, what happens along that line, so limit x y

tends to 0, 0 along y is equal to x.

So, in this case, we are asking for a limit as x tends to 0, where you put y is equal

to x in the in this expression, so you get x cubed minus x cube are divided by 2x square to the power 2. So, this is again 0. So, this is like limit of x tends to 0 of the function

0 is again 0. So, I hope this this is a, it is clear what we what we mean by the limit

along a curve with this example in mind. So, here are the 3 curves for which we computed

the limit. So, along two of them it is 0 and along one of them it does not exist fine.

So, let f be a So, now what is the connection, this is a real point and this is the most

important slide in this video, what is the connection between the limits of a function

along curves and the limit of a function? So, the limit of a function we saw was you

take any sequence tending to that point A tilde and look at F of A and tilde does it

convert somewhere.

So, whereas for a curve we are saying you take only sequences on that curve. So, now

again let f be a scalar valued multivariable function or usual hypothesis. So, the limit

of f at a tilde will exists and equals L precisely when which means, if this happens then whatever is next happens and if that happens then this limit exists and equals L, precisely when

for every curves C in the domain D passing through a tilde the limit of f at a tilde

along C exists and equals L.

So, what are we saying? We are saying that if the limit at a point is L, then whatever

curve you choose the limit along that curve for that function will also be L the other

side if you if for every curve C, as you come close to this point a tilde along that curve

the function value comes close to L, the limit is L and this happens for every curve is the

same, then the limit of f itself not along the curve, but globally is L.

So, this is a very important statement. And the spirit of this statement is for one variable

calculus, we did the same thing when we defined limits. In fact, that was the definition we

said you come along the curve from the right you come from the left and if they match that

was left limit and right limit, then then that is, then whatever value that is that is the limit. So, here we are we have many directions. We have many directions, in fact

more than directions, we have many curves.

So, you look at all possible curves, via which you can come to that point. So, you trace

some curve come to this point ask does the function value tend somewhere. You trace some

other curve come to that point asking does the function value 10 to that same something,

if for any one of those it does not exist, that is it, you that means the global function

value, that limit is not defined. If for all of them, it is some number, it is the same

number, then the global limit value is that same number.

So, important point is, this is often used to show that a limit at a point does not exist.

So, we have seen an example already when we did that example, x cubed minus x squared

y divided by x square plus y square whole square.

Let us do some more examples. So, this is limit of the function x squared minus y squared by x squared plus y squared xy tends to 0, 0. As you can see, these are all rational

functions, wherein you cannot substitute. If you can substitute and get away, go ahead

and do that. That is the best possible scenario. Unfortunately, here, you cannot do that. So,

we have to really sit and compute. So, what we will do is we will again find what is the

limit along the x axis and the y axis.

So, along the x axis, so along the x axis means y is 0, then this becomes limit as x

tends to 0 x squared by x squared, which is clearly 1 along the y axis, this becomes limit

as y tends to 0. So, here x is 1, x is 0. So, you get minus y squared by y squared,

which is minus 1 and already these two values do not match. These values do not match and

because these two values do not match, this limit, does not exist.

How do I know that? Because had the limit existed, then along every curve, you would

have had the same limit. And then that that means in particular along the x axis, you would have had that limit along the y axis also, you would have had that limit, but then

these values do not match. So, I hope it is clear how we use the theorem on the previous

slide.

Let us look at this example, where you have limit xy tends to 0, 0 x y by x squared plus

y squared. Let us do along the x axis, what happens. So, along the x axis, when here you

have to substitute you have y0, so that means you get 0 by x squared plus 0 squared, which

is 0, the symmetric, so as you can see about the y axis also the same thing will happen.

So, limit as y tends to 0, 0 by 0 squared plus y squared. This is 0.

So, unlike our previous case, where we had, we took the two curves, x axis and y axis,

and they already gave different limits. Here, for the axis they are matching. So, there

is still a chance that this limit exists. So, let us look along the line, y is equal

to x. So, along the line y is equal to x, this is you substitute y is equal to x and then take limit x tends to 0. So, this is x squared y, x squared plus x squared, that

is half. And these do not match. So, this does not match with either of these two. So,

this limit does not exist.

So, I hope it is clear what I am doing, I am somehow choosing curves. So, my first possible

choices will be lines, so easiest curves or lines. So, I am choosing lines along which

these limits do not match. Sometimes what will happen is that for every line, the limit

will match. But then you can somehow be smart and choose some other curve.

So, if you want to take the general line, you should take, for example, in this example,

let us look along the line y is mx. Of course, y is mx covers everything other than the line

x is 0, so that we have to make a special case. So, let us see what happens for those,

along the line y is equal mx, well you substitute y is equal to mx. So, that means you get x

times mx squared divided by x squared plus mx to the power 4.

So, this is a mx cube by x squared times 1 plus m squared x squared. And after cancelling,

you can see that this limit is 0. So, this limit does exist and is 0, what happens along

the line x is 0. So, along the line x is 0, you have to substitute x is 0. So, this limit

clearly is 0. So, this is 0 by y to the power 4 is 0. So, you can see that for all the lines,

you are getting the same limit. So, this limit really has a very strong chance of existing.

And in the tutorial, you will be discussing this in more detail. So, I leave this for

now, fine. So, I hope it is clear how to use that previous theorem about the limit of a

function along a curve. It is a very important idea. And, and it is very useful in examples

like this.

So, let us go on to talk about the continuity of a function. So, we have talked about the

continuity for one variable functions using limits of functions. And now we are going

to do the exactly the same thing for multivariable functions. Let f be a multivariable function defined in our domain D in Rk. And a tilde be a point in D. So, now we want a tilde to

be a point in D because we want f to be defined on a tilde such that there exists a sequence

in D which converges to a tilde.

We still want that condition, f is continuous at a tilde if the limit of f at a tilde exist.

And not does it exist, the value of that limit is f of tilde this is what the definition

was even when we add a one variable function. So, f is continuous at a tilde is equivalent

to saying that, if you take a sequence which is convergent to a tilde, then f of an tilde

is convergent to f of a tilde. That is exactly what it means for f to be continuous at a

tilde.

And note what this means note that continuity means the limit at a tilde can be obtained

by evaluating the function at a tilde. So, you can, you can just say that limit is equal

to f of a tilde. So, you can evaluate the function in a tilde. So, the function f is said to be continuous. So, this is all talking about the when the function is continuous

at a point a tilde. So, now we say that the function f the entire function is continuous

if it is continuous at all points of in the domain. So, that means limit x tends x tilde

tends to a tilde f of x tilde is a tilde for all points a tilde in the domain. Fine.

So, let us end with this example of our good old friend x cubed minus y squared x by x

squared plus y squared squared. This is a f x y is not equal to 0, 0 and it 0 if xy

is equal to 00. So, I think you already know what is coming, but let us complete this.

So, the question is this continuous at 0, 0 or is this a continuous function in general?

So, if a tilde is not equal to 0, 0, then limit x y tends to a tilde g of xy. So, if

a tilde is not 0, 0 that means at least one of its coordinates is non-zero say a tilde

a comma b. And that means, if it is not 0, 0 that means at least one of them is not 0.

So, I can write g of x comma y as a quotient. So, let us say I can write it as f of xy by

h of xy both of these are polynomials. So, for polynomials to get the limit I just substitute

and in this case limit as xy tends to a tilde h of xy. This is a h of a tilde and h of a

tilde is not 0. Why is h of a tilde non-zero, because this is going to be equal to a squared

plus b squared whole squared and if ab is not 0, 0, then at least one of them is not

0 that means, this number is non-zero. So, this is not 0.

So, I can use my quotient rule and I can just substitute. So, this is just g of a tilde.

So, this limit exists. So, if a tilde is not 0, 0 the function is continuous at a tilde.

So, the only question is what happens at 0, 0. And we have actually seen what happens,

but I just recall for you. So, limit xy tends to 0, 0 g of xy we checked in a slide a few

minutes ago that this does not exist. So, it just does not exist. Why, because we came

along the x axis and the y axis and along one of the axes, we found that the limit does not exist.

So, therefore in particular, it does not equal g of 0, 0, which is 0 which does not exist,

so there is no question of (())(28:21) this. So, therefore, the function g of xy is not

continuous, or maybe we will see is continuous at all points

except 0, 0. So, I hope this example also clarifies the notion of continuity. This is

really the same as we have done for one variable calculus, except that you have more variables

now. Thank you.